

85. AM I DOING BETTER? A SIMPLE WAY TO VISUALIZE CYCLING PERFORMANCE

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June 19, 2025



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In other sports like running and swimming the competition distance is fixed (for example 5k, 10k, marathon), so it is very simple to quantify and compare performance just using the finishing time of each race. If you do 2min less on a marathon, you have improved - quite simple.

On the other hand, in cycling every time the conditions (distance, elevation gain) of the whole race and/or of specific all-out efforts done within a race, are different and so it is not possible to use just 'time' to cover a certain distance as a performance marker. Of course, power output (average power or normalized power if the effort is stochastic) can give an idea about the level of a certain performance. However, given that each time the duration of maximal efforts is different (unless you are doing a standardized test on a fixed duration, 5min all-out for example), it is not straightforward to evaluate a specific performance. Are you performing better

or worse than in the past? For example, **if a cyclist performs an all-out effort of - let's say - 17min35sec duration and he never performed a maximal effort of that specific duration in the past, how can we evaluate his performance?**

I just started to use and wanted to share this method. It is using the vertical distance from the power-time curvilinear relationship from the specific effort considered.

To see and use this methods you can download the excel file with calculations and explanations here!

Cycling Performance Viewer

70.1KB · XLSX file

[Download](#)

I think this method can give information particularly on individual time trials as you are quite sure that you are looking at maximal efforts for the specific duration. Of course TTs are about speed and not only power, so you should also consider if some variation in aerodynamics has been made when evaluating a given performance.

Additional information that can give context because they influence performance are:

- **Environmental temperature:** low ($<10^{\circ}\text{C}$) and high ($>25^{\circ}\text{C}$) temperature can decrease performance.

[Learn more here.](#)

- **Altitude:** starting from 1500m above sea level, performance can decrease. [Learn more here.](#)
- **Durability:** if the performance have been recorded in the second part of a hard training session/race, it is quite normal that will be lower than in a fresh state. To learn more about durability, just type 'durability' on the search bar of knowledgeiswatt substack home page and read all the articles on the topic.
- **Recovery:** if the performance is recorded after some hard consecutive training/race days (for example, in the last week of a grand tour), it is usually lower than in a fresh state. How much lower? It depends on the recovery capacity of the athlete.

[Learn more here on this topic.](#)

- **Flat vs Uphill Terrain:** 6-7% slope seems the optimal road gradient to obtain maximal power output over 1-20 min duration for both climbers and flat specialists.

[Learn more here.](#)

Thanks for Reading! Share with your Team!

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